

Protocol	service / purpose	Port no.
FTP	Transfers files between computers	20, 21
SSH	Secure remote login and command execution	22
Telnet	Remote login	23
SMTP	Simple Mail Transfer Protocol	25
DNS	Resolves domain names to IP addresses	53
HTTP	HyperText Transfer Protocol	80
POP3	Retrieves / downloads email from a mail server	110
IMAP	Accesses and manages email on the mail server	143
HTTPS	Secure web browsing using SSL/TLS	443
SMB	File and printer sharing over a network	445

* Why open ports = attack surface?

⇒ An open port is considered part of a system's attack surface because it allows network communication with a service running on that port. If the service has a vulnerability, weak configuration, or improper authentication an attacker may attempt to exploit it.

* How to identify open ports? Using which tool

⇒

Open ports can be identified by using port-scanning tools that check whether a TCP or UDP port on a host is accepting network communication. Tool such as Nmap

Nmap (Network Mapper) is a commonly used network-scanning tool.

* Define OSI layers base on the following concept
networking devices

Protocols

Attacks

Port numbers

⇒

1. Physical Layer - Layer 1

→ The Physical Layer is responsible for transmitting raw bits (0s and 1s) through physical media such as copper cables, fiber-optic cables

→ Devices: Hub, repeater, cables, connectors

→ Technologies: Ethernet Physical standards, fiber, radio

→ Attack: cable tapping, signal jamming, physical damage

2. Data Link Layer (2)

→ The Data Link Layer provides communication between devices on the same local network using frames and MAC addresses.

→ Devices: switch, bridge, network interface (NIC)

→ Protocols/technologies: Ethernet, ARP, VLAN, Wi-Fi

→ Attacks: ARP spoofing, MAC flooding, VLAN attacks

3. network Layer - 3L

- The network layer handles logical addressing and routing of packets between different networks.
- Devices: Router, L3 switch
- Protocols: IPv4, IPv6, ICMP
- Attack: IP spoofing, route manipulation

4. Transport Layer

- The transport layer provides end-to-end communication between applications.
- Devices: Firewall, Layer-4 load balancer
- Protocols: TCP, UDP
- Attacks: TCP SYN flood, UDP flood
- Ports: 22, 23, 25, 53, 80, 443, 3389

5. Session Layer

- The session layer establishes, manages and terminates communication sessions between applications.
- Devices: Gateway, session-aware network devices
- Technologies / Protocols: NetBIOS, RPC, SIP
- Attack: session hijacking, session fixation

6. Presentation Layer

- The presentation layer is responsible for data representation, translation, encryption/decryption, and compression.
- Devices: Gateway
- Technologies: TLS/SSL, JPEG, PNG, ASCII, Unicode
- Attacks: TLS downgrades, encryption-related attacks.

7. Application Layer

→ The application layer provides network services directly used by applications.

→ Devices: Proxy server, application gateway

→ Protocols: HTTP, HTTPS, DNS, SMTP, FTP, SSH

→ Attacks: Phishing, HTTP injection, application

→ ports: 80, 443, 53, 25, 22, 22

* Define the key differences between IPv4 and IPv6.

⇒ IPv4 (Internet Protocol Version 4) and IPv6 (Internet Protocol Version 6) are network-layer protocols used to provide logical addressing and route packets across networks. The main difference is that IPv4 uses 32-bit address, while IPv6 uses 128-bit address.

Feature	IPv4	IPv6
Address size	32-bit	128-bit
Address format	Decimal, separated by dots	Hexadecimal, separated by colons
Example	192.168.20.283	2002:db::2
number of address	~ 4.3 billion	~ 3.6×10^{38}
address notation	Four decimal octets	Eight hexadecimal
Header	20-60 bytes	40 bytes
Broadcast	supports broadcast	uses multicast
configuration	manual or DHCP	SLAAC, DHCPv6
NAT	commonly used	generally not
resolution	ARP	NDP

* Define MAC Address.

⇒ A MAC (Media Access Control) address is a unique hardware address assigned to a network interface such as a network interface card (NIC). It is used at the Data Link Layer (Layer 2) of the OSI model to identify a device/interface within a local network.

* What are the possible attacks that can be performed using open ports?

⇒ An open port itself is not an attack. It means a network service is listening and potentially reachable. If that service is vulnerable, misconfigured, or poorly secured, it can become an attack surface.

→ Possible attacks through open ports

→ 21 - FTP : Brute-force login, anonymous access abuse, vulnerable FTP exploitation

→ 22 - SSH : Password brute force, stolen credentials, vulnerable SSH service exploitation

→ 23 - Telnet : credential sniffing, brute force, unauthorized access

→ 25 - SMTP : open-relay abuse, user enumeration, vulnerable mail-server attacks

→ 53 - DNS : DNS enumeration, cache-related attacks, misconfiguration abuse

→ 80 - HTTP : web vulnerabilities such as SQL injection, XSS, file inclusion, authentication attacks

→ 443 - HTTPS : same kinds of web attacks as HTTP, plus TLS/configuration weaknesses

- 445 - SMB : Unauthorized file sharing, credential attacks, vulnerable SMB exploitation
- 3306 - MySQL : Password attacks, unauthorized database access, vulnerable DB exploitation
- 3389 - RDP : Brute force, stolen-credential access, vulnerable RDP exploitation
- 8080 - HTTP Proxy/web app : web attacks, exposed admin, panels, proxy abuse

* Different Between cookies and sessions

Feature	cookies	sessions
stored where?	on the client/browser	mainly on the server
Data location	User's device	server
Security	less secure because data is stored browser, client-side	Generally more secure for sensitive data
size	small, usually around 4KB per cookie	can store more data
Lifetime	can be temporary or have an expiry date	usually expires after timeout or when session is delete.
sent with request?	Browser sends matching cookies to the server	Browser usually sends a session id and server finds the session data
Example	username = Vishul	session-id = abc223

* networking Devices :-

=> Router
gateway
switch
Hub
Modem
Bridge

* topologies

Bus, tree, star, Ring, Hybrid, mesh, Point-to-Point

* communication types

Simplex
Half-duplex
Full-duplex

* Quick comparison Matrix :-

Feature	Analog Transmission	Digital Transmission
Signal form	continuous sine wave	Discrete square wave
Data units	Voltage, frequencies	Bits (0 and 1)
Band width	consumes less raw bandwidth	Demands higher bandwidth
Error correction	not naturally possible	high efficient
Hardware costs	Lower complexity other	High initial

* TCP 3-way Handshake:-

1. SYN (synchronize)

→ Think of it as one party extending their hand for a handshake

→ The process starts with client sending a TCP segment to the server with the SYN Flag set

2. SYN-ACK (Synchronize Acknowledge)

→ This is like the second party tucking the hand

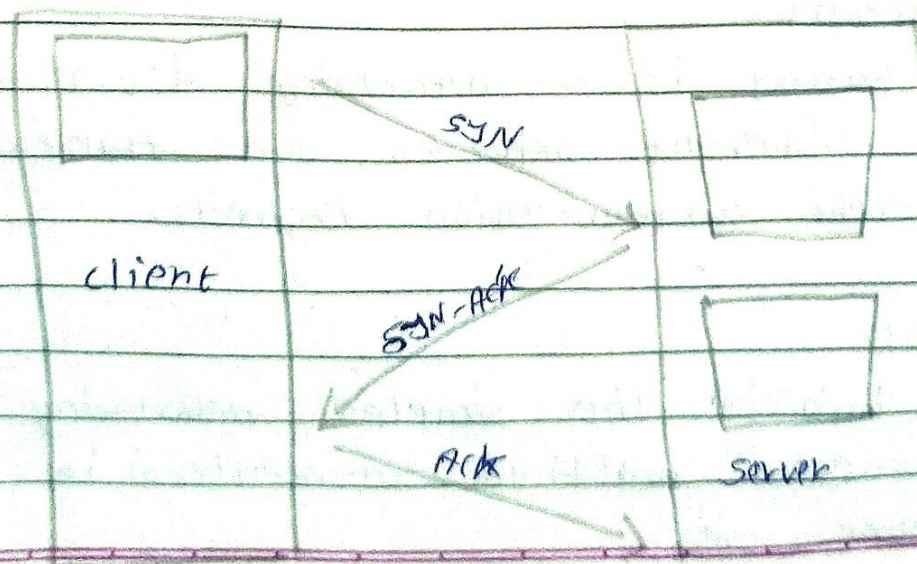
→ The server receives the SYN segment and responds by sending a SYN-ACK segment to the client

3. ACK (Acknowledge):-

→ This is where the two parties shake hands

→ The client receives the SYN ACK segment

→ The client acknowledges the server's response by sending another TCP segment with the ACK Flag set



* Networking Devices :-

2. Bridge :-

→ A bridge is layer 2 networking device that connects two or more LAN segments into a single network. It forwards data frames based on MAC addresses.

2. Router :-

→ A Router is a layer 3 device that connects different network and forwards data packets between them using IP address. It determines the best path for data to travel between networks such as a LAN and the Internet.

3. Modem :-

→ A modem is a device that connects a home or office network to an internet service provider by converting the ISP's transmission signal into digital data.

4. Gateway

→ A gateway is a networking device that connects two different networks and translates between different communication protocols.

5. Hub :-

→ A hub is the simplest networking device it connects multiple computers in the same LAN.

* Different between Private IP and Public IP

Private IP	Public IP
→ Used inside a local network	→ Used on the Internet
→ not directly reachable from the Internet	→ can be reachable from the Internet
→ Used by PCs, Phones, Printers etc.	→ Usually assigned to a router / server
→ Example: 192.168.1.20	→ Example: 8.8.8.8
→ Not unique globally	→ Globally unique

* What are the ranges of Private and Public IP

⇒ Private IP Ranges (IPv4)

- 10.0.0.0 - 10.255.255.255
- 172.16.0.0 - 172.31.255.255
- 192.168.0.0 - 192.168.255.255

Public IP Range

- All IPv4 Addresses except reserved/Private ranges are generally public IP.
- Public IPv4: 2.0.0.0 - 223.255.255.255

* Define different classes of IP address

class	Range	default Mask	Use
A	1-126	255.0.0.0	Larger networks
B	128-191	255.255.0.0	Medium networks
C	192-233	255.255.255.0	Small networks
D	224-239	-	Multicasting
E	240-255	-	Experimental / Reserved

→ 224.x.x.x is reserved for multicast

* What are Port numbers and what are their ranges
=>

A Port number identifies a specific network service/application on a device

Range	Type	Use
0-1023	Well-known Ports	Common services
1024-49151	Registered Ports	Applications/services
49152-65535	Dynamic/Private Ports	Temporary/client connections

total ports numbers 65536

* What is subnetting? what is CIDR /26 and /24?

=> Subnetting: is the process of dividing one larger IP network into smaller networks (subnets).

Ex.

292.268.2.0 /24 → can be divided into smaller networks such as /26.

CIDR

classless Inter-Domain Routing uses /number to indicate how many bits are used for the network portion of an IP address.

/26 → 255.255.0.0

→ 26 network bits

→ 26 host bits

→ more hosts

/24 → 255.255.255.0

→ 24 network bits

→ 8 host bits

→ fewer hosts

/26 → bigger network

/24 → smaller network

* what are type of dns records

⇒ DNS records are instructions in DNS servers that give info about a domain name and map it to IP addresses or services.

types:-

A : Maps your domain name to an IPv4 address.

AAAA: maps your domain name to an IPv6 address.

CNAME: Points an alias or subdomain to a different canonical domain name.

MX: Directs email messages to the correct mail server.

NS: specifies which nameserver is authoritative for the domain.

TXT: Holds text data often for security checks like SPF: or domain ownership proof

SOA: stores core admin details about a DNS zone like refresh rates and contact emails.

PTR: Enables reverse DNS lookups by mapping an IP address back to a domain name.

SRV: defines a specific port and target host for specialized network services.

* Demonstrate how data packets transfer from a server to a client by OSI model.

⇒ When a server sends data to a client, the data passes from Layer 7 down to Layer 1 at the server, travels through the network, and then goes from Layer 1 up to Layer 7 at the client.

OSI

Server → Client

7. Application :- create the data, HTTP response
6. Presentation :- data is formatted/encrypted if needed
5. Session :- manages the communication session
4. Transport :- Adds TCP/UDP information → segment
3. Network :- Adds source/destination IP → Packet
2. Data Link :- Adds MAC addresses → frame
1. Physical :- converts frame into bits/signals and sends it

At the client, the process is reversed.

Bits → frame → Packet → Segment → Data → Application

* what is cybersecurity

=> cybersecurity is the practice of protecting computers, networks, system applications and data from unauthorized access, attacks, or theft

-> cybersecurity means protecting digital systems and information from cyber threats

* The main goals are call the CIA Triad.

=> C :- confidentiality

-> only authorized people can access information

I :- integrity

-> information should not be changed or manipulated without permission

A :- Availability

-> systems and data should be available when authorized user need them

* Example of cyber threats:-

- malware and viruses
- phishing
- Ransomware
- Password attacks
- DDoS attack
- Unauthorized access
- Data theft

* What is firewall?

⇒ A firewall is a network security system that monitors and controls incoming and outgoing network traffic according to predefined security rules.

→ firewall trusted ~~over~~ untrusted network
keeps ~~se~~ to security barrier provide

* Target / objectives of firewall

⇒ Prevent unauthorized access

- block malicious or unwanted traffic
- monitor network traffic
- protect internal system and data
- maintain logs of network activity
- allow legitimate traffic according to security rules.

* statefull inspection firewall

- ⇒ A statefull inspection firewall keeps track of the state of active network connections.
- for example, if a computer sends a request to a web server, the firewall remembers that connection and allows the valid response back.

* communication types:-

1. Simplex:-

- direction :- Unidirectional
- How it works:- one device sends data and the other device only receives it the receiver cannot reply back through the same channel

2. Half duplex:-

- Direction :- Bidirectional, but alternate
- How it works :- Both devices can send and receive data, but they must take turns while one device transmits the other can only listen

3. Full - Duplex:-

- Direction :- Bidirectional and simultaneous
- How it works :- Both device connected can send receive data at the exact same moment without waiting for each other.